# Monte Carlo Simulation of The 2D Ising Model

KINETIC THEORY AND STOCHASTIC SIMULATIONS


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# 1 Introduction

This research focuses on studying the two-dimensional Ising model, which serves as a representation of spin interactions on a two-dimensional lattice. The lattice comprises spins that can be oriented either upwards or downwards. When the temperature is low, the system demonstrates a significant level of organization known as **ferromagnetism**, where most spins align in the same direction. However, when the temperature exceeds a critical threshold (Tc), the ordered arrangement of spins is disrupted, leading to a **paramagnetic** state where spin alignment is no longer favored. The investigation aims to observe the system's temporal evolution while considering specific variables such as temperature, energy, and magnetization. Monte Carlo simulations are employed to simulate this process and estimate the probabilities of different outcomes. This is necessary because random variables are involved, making it challenging to predict the precise behavior of the system.

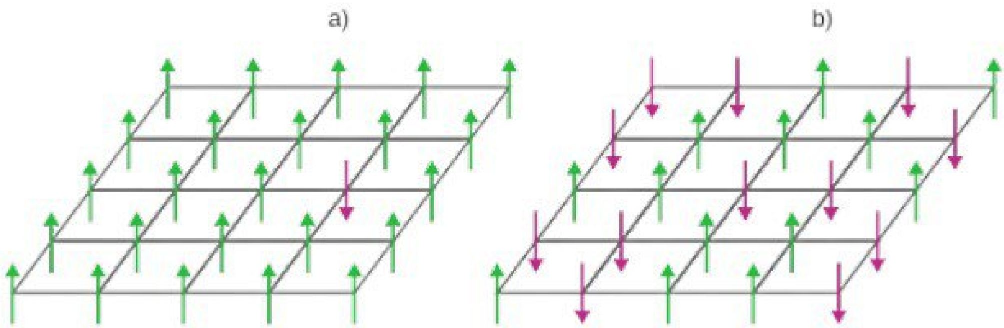



**Figure** 1: The 2-D Ising Model represents spin orientations, with arrows indicating either spin up or spin down. In (a), the model depicts the Ferromagnetic state, characterized by an ordered arrangement of spins, when the system is below the critical temperature (Tc). In contrast, (b) illustrates the paramagnetic state, where the system is above Tc and lacks order

## 1.1 Defintion of Terms

**Magnetic Moment:** The magnetic moment of a magnet is a characteristic that governs how the magnet reacts to an external magnetic field. It is a vector property, possessing both magnitude and direction, and it points from the south pole to the north pole of the magnet. The magnitude of the magnetic field created by a magnet is directly proportional to its magnetic moment. Essentially, the magnetic moment reflects the magnet's capacity to generate a mechanical force when it interacts with a magnetic field that is applied to it.

**Ferromagnetism:** Certain materials, including iron, cobalt, nickel, and their alloys, exhibit a fundamental property known as ferromagnetism. This property allows these materials to either become permanent magnets themselves or be attracted to magnetic fields. In ferromagnetic substances, the spins of individual particles align together, leading to magnetic moments that all point in the same direction.

**Paramagnetism:** Paramagnetism is a type of magnetism observed in specific materials. When these materials are exposed to an external magnetic field, they are attracted towards the direction of that field. These materials also generate an internal magnetic field, although it is relatively weaker than the applied field.

# 2 Theory

## 2.1 Square-lattice 2D Ising Model

Consider a square lattice $\Lambda$ of linear size L, with periodic boundary conditions on the horizontal and the vertical directions. Your task is to write a numerical code based on the Metropolis Monte Carlo algorithm to simulate the dynamics of the nearest-neighbor ferromagnetic Ising model with hamiltonian;

$$H(\sigma) = -J \sum_{<i,j>} \sigma_i \sigma_j, i, j \in \Lambda, J > 0 \tag{1}$$

where $\sigma = (\sigma_i)_{i \in \Lambda}$ denotes a configuration of spins on $\Lambda$, with $\sigma_i \in \{-1, +1\}$, and where the sum is taken over pairs of adjacent spins (recall that each pair must be counted only once!). In the thermodynamic limit, the 2D Ising model undergoes a phase transition at the inverse critical temperature:

$$\beta_c = \frac{\ln(1+\sqrt{2})}{2J} \tag{2}$$

For $\beta > \beta_c$ the 2D Ising model exhibits a spontaneous magnetization

$$m_\beta = \left[1 - \left(\frac{1}{\sinh^4(2\beta J)}\right)\right]^{\frac{1}{8}} \tag{3}$$

In your simulations set $k_\beta = J = 1$ (in this case $T_c = \beta_c^{-1} \approx 2.269$) and $L = 100$. Time is measured in number of sweeps $N = L^2$.

*Initial configurations.* For all $i \in \Lambda$ take

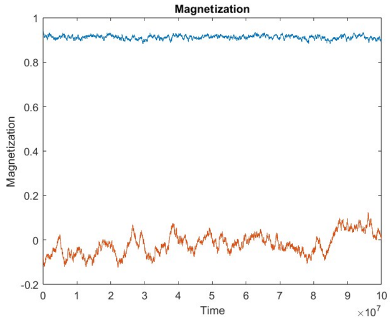
1. $\sigma_i = +1$

2. $\sigma_i = -1$

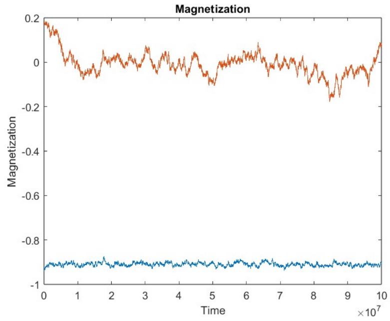
3. $\sigma_i = +1$ with probability 1/2

# 3 Numerical Results and Discussion

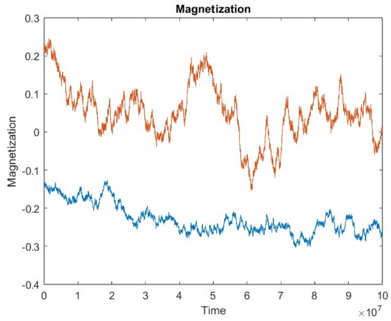
## 3.1 The Magnetization of the System




(a) Fig. 2: Positive Interaction



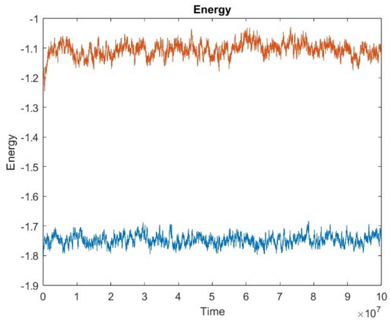

(b) Fig. 3: Negative Interaction



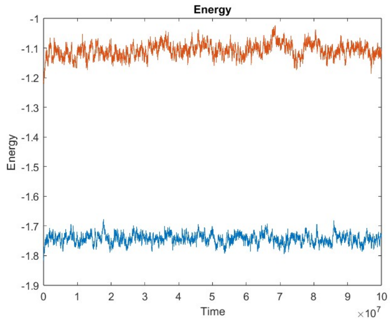

(c) Fig. 4: Random Interaction

A positive spin led to a magnetization of **10000**, indicating a strengthened magnetic field within the material. In contrast, a negative spin resulted in a magnetization value of **-10000**, indicating an anti-parallel arrangement of spins in the lattice. On the other hand, the random spin generated a magnetization value of **136**, indicating a negative interaction at that particular random value.

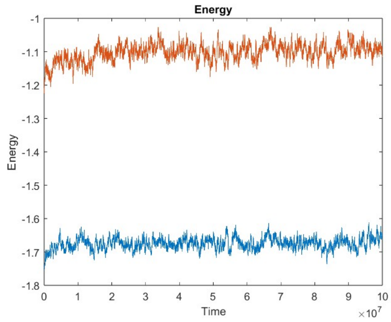
## 3.2 The Energy of the System




(a) Fig. 5: Positive Interaction



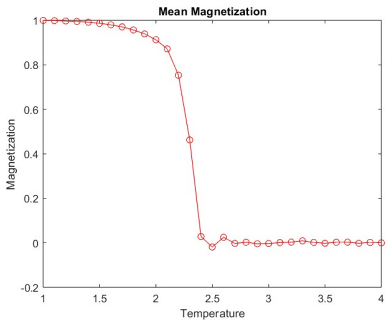

(b) Fig. 6: Negative Interaction



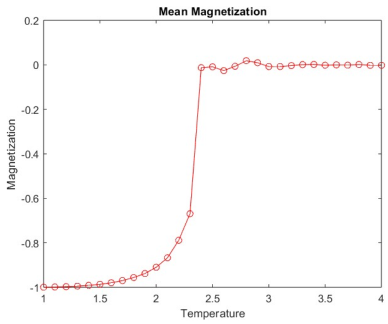

(c) Fig. 7: Random Interaction

For both positive and negative interactions, the total energy of the system was calculated to be **-20000**. However, in the case of random interactions, the energy was determined to be **-268**, indicating that the particles were confined within the limited distance of the lattice square. The random interaction displayed diverse values, ranging from 5, implying that the energy input to or extraction from the system could fluctuate based on the level of interaction.

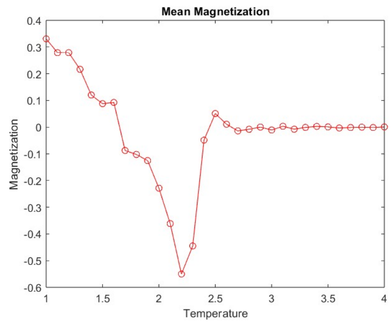
### 3.3 The Mean Magnetization of the System




(a) Fig. 8: Positive Interaction



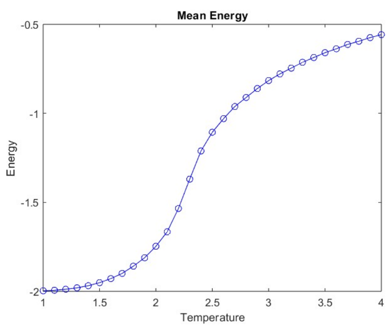

(b) Fig. 9: Negative Interaction



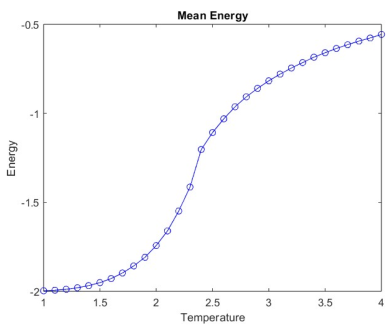

(c) Fig. 10: Random Interaction

The positive interaction exhibited an average magnetization of **1**, indicating that a substantial number of spins aligned in the same direction as the applied magnetic field. Conversely, the negative interaction resulted in an average magnetization of **-1**, implying a predominant alignment of spins in the opposite direction of the field. In the case of the random interaction, the average magnetization was **0.0136**, indicating a near-zero alignment of spins with the applied field on average. The positive mean magnetization suggests that as the field intensity increases, a larger number of spins flip and align with the field.

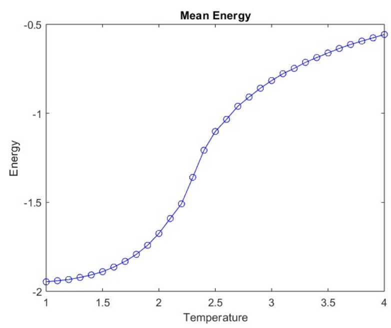
### 3.4 The Mean Energy of the System




(a) Fig. 11: Positive Interaction



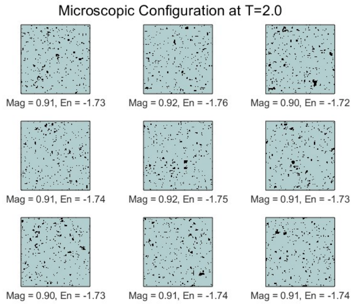

(b) Fig. 12: Negative Interaction



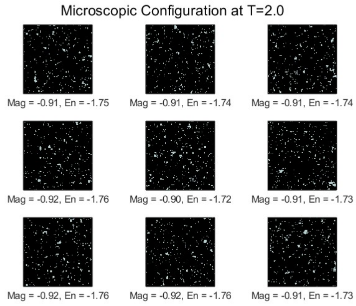

(c) Fig. 13: Random Interaction

The mean energy of the system was determined to be -2 for both positive and negative interactions. This observation can be attributed to the introduction of energy into the system, which enables the flipping of spins.

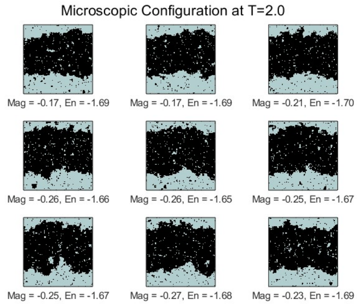
### 3.5 The Microscopic Configuration at T = 2.0




(a) Fig. 14: Positive Interaction



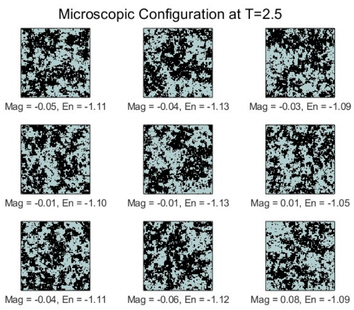

(b) Fig. 15: Negative Interaction



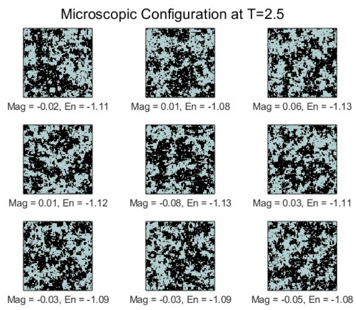

(c) Fig. 16: Random Interaction

At a temperature of 2.0, the positive interaction configuration exhibits predominantly blue regions, unlike the negative and random interactions. This occurrence can be attributed to the limited flipping of spins in the positive interaction at low temperatures. These findings suggest the presence of ferromagnetic properties in the material. In contrast, the results for both the negative and random spins demonstrate distinct behaviors.

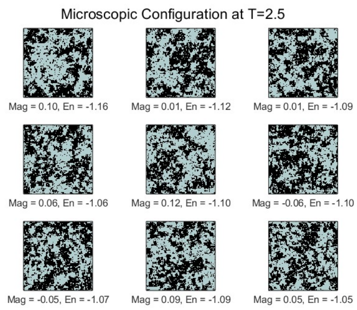
### 3.6 The Microscopic Configuration at T = 2.5




(a) Fig. 17: Positive Interaction



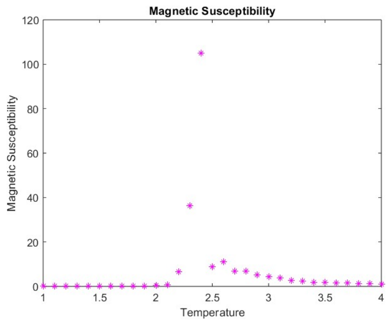

(b) Fig. 18: Negative Interaction



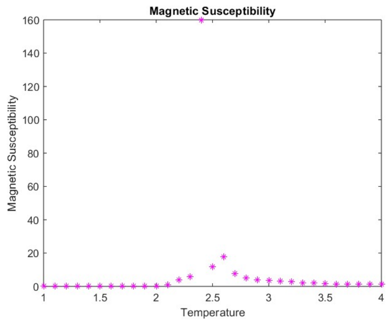

(c) Fig. 19: Random Interaction

When the temperature is T=2.5, there is an increased probability of spin flipping, resulting in a decrease in magnetization within the material. The spins no longer align, and the resulting configurations appear random. As a result, the material exhibits paramagnetic behavior.

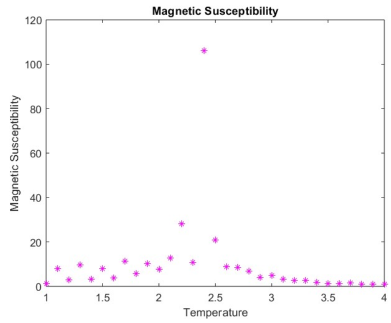
### 3.7 The Magnetic Susceptibility of the System




(a) Fig. 20: Positive Interaction

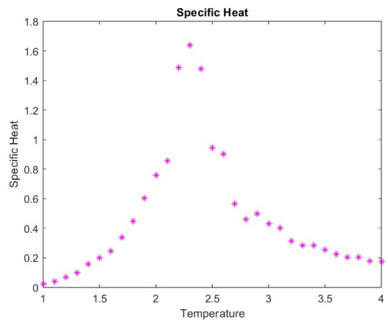



(b) Fig. 21: Negative Interaction

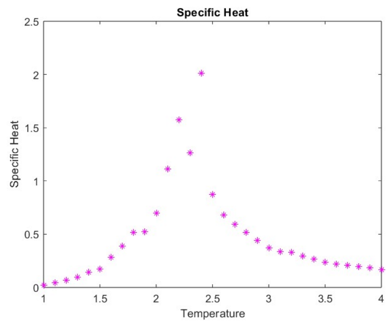



(c) Fig. 22: Random Interaction

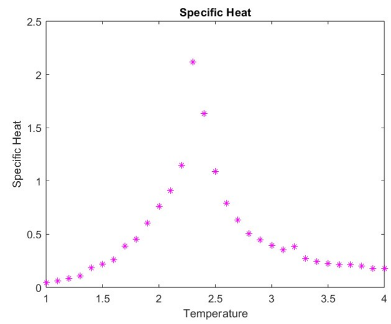
### 3.8 The Specific Heat of the System




(a) Fig. 23: Positive Interaction




(b) Fig. 24: Negative Interaction




(c) Fig. 25: Random Interaction

# 4 Appendix

## 4.1 MatLab Codes for Monte Carlo Simulations

### 4.1.1 Intialization

```
% Here is the MATLAB code
function [spin, Mag] = Initialization(L, IC)
% INITIALIZATION ALGORITHM
if IC == 1
    spin = ones(L, L);
elseif IC == 2
    spin = -ones(L, L);
else
    spin = sign(0.5 - rand(L, L));
end
global Mag;
global En;
global EngyMean;
global MagneticMean;
En = 0;
Mag = 0;
for x = 1:L
    for y = 1:L
        N = Neighbor(L, x, y);
        Mag = Mag + spin(x, y); % Initial magnetization
```

```
        En = En - spin(x, y) * (spin(N(4), y) + spin(x, N(2))); % Initial energy
    end
end
En
Mag
EngyMean = En / L^2
MagneticMean = Mag / L^2
end
```

Listing 1: Initialization

### 4.1.2 Neighbor

```
% Here is the MATLAB code
%The purpose of this script is to remove most of the unnecessary distance
% calculations by monitoring the spins that are in close proximity to each %other.
function [Neighbors] = Neighbor(L, x, y)
above = mod(x - 2, L) + 1;
below = mod(x, L) + 1;
left = mod(y - 2, L) + 1;
right = mod(y, L) + 1;
Neighbors = [above, right, left, below];
end
```

Listing 2: Neighbor

#### 4.1.3 Metropolis

```
% MATLAB code goes here
function [spin] = Metropolis(spin, T, En, Mag, L, n, EngyMean, MagneticMean)
% METROPOLIS function performs the Metropolis algorithm for spin updates
global En;
global Mag;
global EngyMean;
global MagneticMean;
global En2Mean;
global Mag2Mean;
global Check;
j = 1;
E = 0;
M = 0;
En2 = 0;
Mag2 = 0;
y = 1;

for i = 1:n
    % Randomly select a spin to update
```

```
    linearIndex = randi(numel(spin));
    [row, col] = ind2sub(size(spin), linearIndex);
    % Calculate the energy change (dE) due to the spin flip
    N = Neighbor(L, row, col);
    dE = 2 * spin(row, col) * (spin(N(1), col) + spin(row, N(2)) +
       spin(row, N(3)) + spin(N(4), col));
    if dE >= 0
        prob = exp(-dE / T);
        x = rand();
        if x <= prob
            % Accept the spin flip
            spin(row, col) = -spin(row, col);
            En = En + dE;
            Mag = Mag + 2 * spin(row, col);
        end
    else
        spin(row, col) = -spin(row, col);
        En = En + dE;
        Mag = Mag + 2 * spin(row, col);
    end
    if (Check == 1)
        % Calculate quantities for checking purposes
        E = E + En / (L^2);
        M = M + Mag / (L^2);
```

```
        En2 = En2 + (En / (L^2))^2;
        Mag2 = Mag2 + (Mag / (L^2))^2;
    end
    if (T == 2) || (T == 2.5) || ((T == 1) && (Check == 0))
        if mod(i, 1000) == 0
            Energy(j) = En / (L^2);
            Magnetization(j) = Mag / (L^2);

            if ((mod(j, 1000) == 0) && (j <= 9000))
                if (T == 2)
                    figure(6)
                    sgtitle('Microscopic Configuration at T=2.0')
                else
                    figure(9)
                    sgtitle('Microscopic Configuration at T=2.5')
                end
                if (y <= 9)
                    subplot(3, 3, y);
                    image((spin + 1) * 100);
                    colormap(autumn);
                    xlabel(sprintf('Mag = %0.2f, En = %0.2f', Mag / L
                        ^2, En / L^2));
                    set(gca, 'YTickLabel', [], 'XTickLabel', []);
                    axis square;
```

```
                    colormap bone;
                    drawnow;
                    y = y + 1;
                else
                    y = 1;
                end
            end
            j = j + 1;
        end
    end
end
if (Check == 1)
    % Calculate mean values
    EngyMean = E / n;
    MagneticMean = M / n;
    En2Mean = En2 / n;
    Mag2Mean = Mag2 / n;
end
if (T == 1) && (Check == 0)
    Time = 1:1:j-1;
    Time = Time * 1000;
    figure(5)
    plot(Time, Magnetization, 'b')
    title('Magnetization at Equilibrium')
```

```
    xlabel('Time')
    ylabel('Magnetization')
end
if (T == 2) || (T == 2.5)
    Time = 1:1:j-1;
    Time = Time * 10000;
    figure(1)
    plot(Time, Energy)
    title('Energy')
    xlabel('Time')
    ylabel('Energy')
    hold on
    figure(2)
    plot(Time, Magnetization)
    title('Magnetization')
    xlabel('Time')
    ylabel('Magnetization')
    hold on
end
end
```

Listing 3: Metropolis

### 4.1.4 Project MainScript

```
clear; clc; close all;
```

```
tic
global MagneticMean;
global EngyMean;
global Mag;
global En;
global En2Mean;
global Mag2Mean;
global Check;
L = input("Enter Lattice Size: ");
IC = input("Enter The initial conditions \n1 = Positive, 2 = Negative
    , 3 = Random\n");
Check = 0;
spin = Initialization(L, IC);
n = 10^8;
spin = Metropolis(spin, 1, En, Mag, L, n, MagneticMean, EngyMean);
n = 10^7;
i = 1;
for T = 1:0.1:4
    Check = 1;
    spin = Metropolis(spin, T, En, Mag, L, n);
    Energy(i) = EngyMean;
    Magnetization(i) = MagneticMean;
    MagSus(i) = ((L^2) / T) * (Mag2Mean - (MagneticMean^2));
    SpesHeat(i) = ((L / T)^2) * (En2Mean - (EngyMean^2));
```

```
    i = i + 1;
end
Temp = 1:0.1:4;
figure(3) % FIGURE 3
plot(Temp, Energy, 'b-o')
title('Mean␣Energy')
xlabel('Temperature')
ylabel('Energy')
figure(4) % FIGURE 4
plot(Temp, Magnetization, 'r-o')
title('Mean␣Magnetization')
xlabel('Temperature')
ylabel('Magnetization')
figure(7) % FIGURE 7
plot(Temp, SpesHeat, 'm*')
title('Specific␣Heat')
xlabel('Temperature')
ylabel('Specific␣Heat')
SpesHeat
figure(8) % FIGURE 8
plot(Temp, MagSus, 'm*')
title('Magnetic␣Susceptibility')
xlabel('Temperature')
ylabel('Magnetic␣Susceptibility')
```

```
MagSus
toc
```

Listing 4: Project original Script